Supplementary material: information bounds, covariance validation and reproducibility

S1. Continuous and discrete likelihood

All parameters share known positive stationary OU covariance. The innovation transformation separates an independent normal initial observation from a Brownian drift experiment. The translation theorem [1] suffices for this common-covariance case; a changing-covariance criterion [2] is unnecessary. The family-wide consistency obstruction is classical [3]. Unknown initial conditions would enter the sensitivity matrix as nuisance parameters.

For the exponential model, set $q_a(t) = x_0(a + \tau^{-1})e^{at}$ and $\nu^2 = 2\sigma^2/\tau$. Continuous likelihood is

$$\log L(a; Y) = \nu^{-2} \left[\int_0^T q_a(t)(dY(t) + Y(t)dt/\tau) - \frac{1}{2} \int_0^T q_a(t)^2 dt \right] + \text{constant}.$$

The initial term is independent of a when x_0 is fixed. The exact discrete negative log likelihood, with $e_n(a) = y_n - x_0 e^{at_n}$, is

$$-\log L_h(a) = \frac{e_0(a)^2}{2\sigma^2} + \frac{\sum_{n=1}^{N} [e_n(a) - \rho e_{n-1}(a)]^2}{2\sigma^2(1-\rho^2)} + \text{constant}.$$

The independent validation constrains optimization to $[-2.5, 0.5]$ before simulation. Sixty-one parameter-grid points identify candidate local minima, each refined by bounded scalar optimization; endpoints are checked. This is a numerical bounded-likelihood estimate, without a proof that all minima between coarse points are found. It need not be unbiased or attain the Cramér–Rao bound. Its squared error is finite by boundedness. The weak ratio is summarized by quantiles rather than a population MSE.

The separate geometry execution performs deterministic sensitivity quadrature for the scalar example. It computes the joint information for (a, x_0), its Schur complement and the weak influence's normalization and initial-state annihilation. Both identities are verified to 10^{-9} on the finest reference grid. Its config, source fingerprint and output archive are retained; this computation adds no stochastic paths and is not an additional Monte Carlo replication.

S2. Mixed-noise covariance

Let each sample error contain OU variance $v_{c,j}$ and independent nugget variance $v_{w,j}$, with total $v_j = v_{c,j} + v_{w,j}$. For assumed correction $\hat{v}$, use $\tilde{f} = \exp[\frac{1}{2} \sum_j (v_j - \hat{v}_j)\partial_j^2] f$ in the residual coefficients. The exact discrete covariance is

$$\Sigma_r = \sum_{\alpha \neq 0} \alpha! \left[v_c^\alpha U_\alpha^T R_{|\alpha|,h} U_\alpha + (v^\alpha - v_c^\alpha) U_\alpha^T U_\alpha \right].$$

Distinct times share only OU covariance; equal times contain total marginal variance. The response–library covariance is included in U_α. Correctness fixtures compare this expression with Gaussian quadrature, including raw and miscalibrated features.

S3. Covariance-estimation Monte Carlo error

For independent centered vectors r_m with covariance Σ, use $S_n = n^{-1} \sum_m r_m r_m^T$ and theoretical centering. Independence gives

$$E\|S_n - \Sigma\|_F^2 = \frac{E\|r\|^4 - \|\Sigma\|_F^2}{n}.$$

For Gaussian vectors $E\|r\|^4 = (\operatorname{tr}\Sigma)^2 + 2\|\Sigma\|_F^2$, making the relative RMS floor

$$\frac{\sqrt{(\operatorname{tr}\Sigma)^2 + \|\Sigma\|_F^2}}{\sqrt{n}\,\|\Sigma\|_F}.$$

Polynomial residuals above order one are non-Gaussian. The protocol estimates their fourth moment and labels the Gaussian expression as a reference. An affine-null bootstrap centers archived residuals, whitens empirical covariance, recolors to the analytical target and resamples 250 multinomial count vectors with the original path count. Covariance eigenvalues are floored at 10^{-12} of the largest during whitening. This approximate null retains empirical shape; its exceedance fraction is not an exact test p-value.

Coefficient validation uses 2,000 paths at amplitudes $0.001, 0.01, 0.05, 0.2$. Errors subtract the clean discrete fit. The residual surrogate uses the true coefficients; the exact finite-grid derivative about the clean fit would instead use its slightly quadrature-shifted coefficients. This remaining quadrature-scale distinction is retained rather than described as an exact finite-grid coefficient identity. A guard requires every support-restricted design's smallest singular value to exceed half the clean value. Actual covariance is conditional on this event, whereas the linearized Gaussian covariance is unguarded. Counts, bias and paired linearization errors accompany discrepancy values. The protocol does not presume finite unconditional nonlinear estimator moments.

S4. Protocol, implementation and provenance

`configs/revision.json` was fixed before the corresponding numerical outputs; it is not a public preregistration. Numerical experiments execute in CPU Kaggle notebooks. Batched deterministic Van der Pol integration uses the recorded solver and tolerances; a correctness check compares batched and individual trajectories. Duplicate executions are not counted as independent evidence. Six synthetic residual arrays are used for covariance-error calibration. Notebook source fingerprints identify the executed implementation; local manuscript edits do not regenerate those execution snapshots.

Seed ranges start at 1,000,000 for sparse regression, 1,100,000 for scalar likelihood, 1,200,000 for coefficient validation, 1,300,000 for covariance bootstrap and 1,500,000 for heterogeneous records.

Other experiment ranges are disjoint. Seed reuse within a condition pairs grids and methods. Conditions use common random numbers and cannot be pooled as independent experiments.

Four systems, two regimes, two amplitudes, four grids, three methods and 128 paths produce 24,576 fits. Thirty fourth-power bump functions have full support width 1.6 for decay and three for oscillators. All centers remain fixed. The correlation ratio is $\tau/L = 0.1$. The independent protocol has no forecasting endpoint. Scalar likelihood has 1,000 paths on six nested grids. Heterogeneous acquisition uses 400 blocks with up to 16 recordings, initial offsets uniform on $[-0.25, 0.25]^2$ around $(2, 0)$ and fixed sensor noise amplitudes derived from the baseline path.

Relative MSTLS uses zero ridge penalty, unit rescaling, 50 logarithmic thresholds 10^{-4} to one and loss weight 0.8. Seven equation fixtures matched the pinned author regression code exactly; the repository commit and SHA-256 are recorded in the parity report. This component does not reproduce adaptive WSINDy or simultaneous noise reconstruction in WmSINDy [4, 5]. Author source is inspected separately and excluded from the distribution package because no redistribution license was found in the inspected tree.

Every output retains source fingerprint, execution environment and completion status. Completed notebooks remain immutable snapshots. Seven correctness tests check exponential covariance application, integration by parts, OU sampling, Hermite correction, exact raw/miscalibrated/mixed covariance, clean recovery and conservative diagnostic refusal. Additional checks verify common-variance scaling of IRLS and batched versus individual ODE integration. Local postprocessing creates tables and figures from downloaded results. Median intervals resample independent paths; support intervals are Wilson intervals. No global multiplicity-adjusted comparison is claimed.

S5. Aggregation and approximation diagnostics

Coefficient-level aggregation diagnostics use the same 400 blocks, random seeds, initial conditions and sensor errors as Table S5. A separate execution preserves the coefficient matrices without changing the estimators. It replays the existing experiment and is not an additional set of independent records. Error norms are checked against the original output before combining these results.

For coefficient error vectors e_m normalized by $\|\xi^*\|_F$, define $B = \|\bar{e}\|$, $S^2 = n^{-1}\sum_m \|e_m - \bar{e}\|^2$ and $R^2 = n^{-1}\sum_m \|e_m\|^2$. The empirical identity $R^2 = B^2 + S^2$ separates squared bias of the sample mean from dispersion; finite-sample B^2 includes sampling error of that mean. An adjusted estimate subtracts $\mathrm{tr}(\widehat{\mathrm{Cov}}e)/n$ using the unbiased sample covariance. Medians alone do not identify this decomposition.

The relative RMS linearization diagnostic in Table S4 divides the RMS difference between actual and linearized coefficient displacements by the RMS linear displacement. It can exceed one. All 2000 paths pass the singular-value guard at each amplitude, so guarded and complete empirical samples coincide; this is not proof of finite population moments.

S6. Complete adaptive author-pipeline reproduction

A separate protocol, `configs/full_adaptive.json`, was frozen before comparative outputs. It uses Van der Pol with parameter one, initial state (2,0), duration 20, noise amplitude five percent, OU correlation time 0.3 or independent errors, and meshes 0.04 and 0.01. Seeds 910000–910015 give 16 independent records per cell. All algorithms receive the same fine-grid error realization and its coarse restriction; SHA256 checks verify paired inputs. This seed range is separate from the earlier studies. Discovery libraries have nine nonconstant monomials of total degree one through

three, matching the released WmSINDy example. Fixed-window discovery and supplied-support comparators are rerun on that library, rather than reusing ten-feature results.

Pinned author repositories are WSINDy_ODE commit `72d8d6e5be2b`, WENDy commit `65c52a893b1f`, and the WmSINDy Van der Pol example at commit `f1a781009a5c` [4–6]. Files or repository ZIPs are fetched from those revisions and verified with the full revision identifiers and SHA256 values in `configs/author_pins.json`. Core author functions are executed without rewriting the statistical algorithms. Author sources are excluded from distributed archives; download instructions and our adapters are provided.

WSINDy uses FFT-corner widths and polynomial degrees, gradient-adapted placement with relax_AG=1, automatic SMAF smoothing initialized as in the released driver, scaling parameter two, 100 thresholds from 1e-4 to one, zero ridge penalty and loss weight 0.8. The adaptive placement is explicitly enabled relative to the released driver's uniform-placement default. GLS remains disabled, as in that driver. Selected widths, degrees, centers, smoothing-filter lengths and losses are saved.

WENDy uses the released set-and-forget settings: estimated noise, the exp(-9/(1-x^2)) bump, automatic minimum radius and four dyadic multiples, SVD corner truncation, maximum 100 iterations, relative tolerance 1e-10, diagonal relaxation 1e-10, and normality checks after iteration ten with cutoff 1e-4. Its four-term structure is prespecified. Actual coefficients are withheld: the diagnostic true_vec is replaced by a unit vector and use_true=0. The returned normality value and iterates are archived. These are the released iid-noise equations; under physical OU error they are evaluated as a misspecified observation model, without claiming a full adaptive OU-WENDy extension.

The MATLAB functions execute under GNU Octave 8.4.0 with statistics and signal packages. Separate compatibility functions provide exact analytic polynomial Jacobians in the declared feature order, exact values and first derivatives of the specified bump with author normalization, and the two-input pagemtimes operation used in covariance construction. These replace Symbolic Toolbox and missing host operations. They do not change radius selection, SVD, covariance updates, regression or stopping logic. Bitwise agreement with a proprietary MATLAB runtime is not asserted.

WmSINDy executes the author Python initializer, RK4 predictions, weak-residual loss, gradient updates, adaptive window function, pruning and least-squares refits. Every completed fit performs eight outer cycles of 5000 Adam steps (40,000 total), with threshold 0.15, learning rate 0.001, epsilon 1e-7, q=1 and decay 0.9. Noise is initialized to zero. Legacy Adam is used with TensorFlow 2.20.0 / tf-keras 2.20.1. Slots for new coefficient variables are created outside traced functions, and the author loss functions are retraced after each window update so changed variable-container shapes are actually used. Adam is initialized independently for each recording; noise-variable slots and iteration count continue through its eight cycles. The released all-valid-convolution-row calculation is retained. Loss checkpoints, eight coefficient iterates, window widths, spectral corners, optimizer counts and the reconstructed noise signal are archived.

The compatibility control uses the author's Van der Pol example: parameter 0.5, initial state (-2,1), duration ten, 1000 linspace points, iid Gaussian amplitude 25 percent and seed one. Its nominal WmSINDy prediction step 0.01 is retained although the linspace spacing is 10/999. The benchmark instead uses its exact grid spacing. Clean trajectories are integrated with DOP853 at relative and absolute tolerances 1e-12 for this control. The single control checks execution and recovery and does not reproduce the published 50-run aggregate figure. Reported comparison coefficient errors use the Frobenius norm.

The comparison contains 448 planned fits: four discovery algorithms and three supplied-support

parameter estimators, four observation cells and 16 records. Discovery success is exact support equality; supplied-support estimators receive no structure-discovery score. Failure counts remain in success denominators. Error medians and bootstrap intervals condition on completed fits; all failures and their logs are retained. Mesh differences are paired by recording. Wilson limits accompany support counts in `reports/full_adaptive_results.json`. Computational runtimes are execution diagnostics, not cross-language speed claims. With one oscillator and 16 paths per cell, these results do not justify universal rankings or fine precision of support probabilities.

S7. Information, window optimization and coefficient replay

The configuration `configs/reviewer.json` specifies supplementary calculations for Sections S9.7–S9.9. The Van der Pol calibration uses the identical clean h=0.01 trajectory and coordinate standard deviations as the complete-pipeline comparison. Noise variances are held fixed under every parameter perturbation. Four independently free coefficients are ordered as velocity in equation one, then position, velocity and position-squared times velocity in equation two. The augmented DOP853 system integrates the two states, eight coefficient sensitivities and four initial-state sensitivities, with relative tolerance 1e-11 and absolute tolerance 1e-13. Stationary OU innovations give exact sampled-trajectory information from these numerical sensitivities; trapezoidal integration evaluates the continuous norm on grids from h=0.04 down to 0.00125. Central coefficient perturbations of size 1e-5 check sensitivities at 101 points. The covariance matrices, joint information, sensitivities and their derivatives are archived.

The scalar window optimum uses the complete cross-window residual covariance and the normalization constraint in Section 4.6. Covariance eigenvalues below 1e-10 or 1e-12 times the largest are discarded. The normalization vector is projected onto the retained eigenspace, where the constrained quadratic optimum is computed. The resulting influences are checked for unit coefficient sensitivity and zero initial-state sensitivity. Quadrature uses 4096 and 8192 intervals. The spectral-cutoff dependence of the enriched 120-window family is retained in S9.8 and numerical report; no global shape optimum is claimed. The calculation uses generating parameters and noise covariance, so the numbers concern an oracle local Gaussian reference.

The heterogeneous coefficient replay exactly reproduces all 3600 method/block/record-count errors of the 400-block aggregation experiment. It saves coefficients in addition to norms and does not supply new independent evidence. Bias and dispersion obey the empirical squared-error identity in S5. The bias-fraction adjustment subtracts the estimated sampling variance of the coefficient sample mean; neither adjusted nor unadjusted fractions are claimed as exact population quantities.

For the 16-path adaptive comparison, relative RMSE uses the complete coefficient matrices, including falsely selected terms, and the same Frobenius denominator as the error medians. Its 95% intervals resample independent records 5000 times. Comparing that RMSE with a local unbiased information reference indicates the scale of remaining error; it does not establish unbiasedness, regularity, calibrated uncertainty or attained efficiency.

S8. Efficiency protocols and complete summaries

Theoretical proofs and their function-space assumptions are stated in Sections 4.3–4.6 and in the standalone proof document. The configuration `configs/efficiency.json` defines two experiments; each executed notebook retains its own exact configuration and source fingerprint. Numerical experiments run on private CPU Kaggle notebooks. Synthetic OU paths use seeds 2100000–2100999 for the scalar experiment and 2200000–2200127 for Van der Pol, independent of other studies.

Noise amplitudes, methods and window families are paired within seed. Resampling uses 5000 independent-path bootstrap draws with generator seed 2302600; paired comparisons resample one common path-index vector. These are unadjusted exploratory intervals.

The scalar cutoff is the product of two quintic smoothstep ramps. For distance d from an endpoint and width δ, $z = \min(1, \max(0, (d - \delta/4)/(3\delta/4)))$ and $\eta = z^3(10 - 15z + 6z^2)$. Its derivative is included analytically in the weak equation. The functions are compactly supported and C^2, which suffices for the implemented weak form; smooth approximation yields the same covariance closure. Scalar experiments use 65536 quadrature intervals. The pilot and every final scalar estimate are clipped to $[-2.5, 0.5]$ for the bounded-risk summaries. Unbounded error medians and every estimate are preserved. The ratio denominator diagnostic uses the generating clean moment only after fitting; it neither rejects observations nor changes estimates.

Table S1. Scalar fitted-pilot summaries for every amplitude and cutoff. RMSE, bias and dispersion use bounded final estimates. All 1000 paths enter each row. Clips count altered final estimates; diagnostic pass counts do not select a subset.

Noise SD	Cutoff	Median	RMSE	Bias	Dispersion	Clips	Diagnostic passes
0.1	0.2	0.067465	0.101006	0.002470	0.100976	0	1000/1000
0.1	0.1	0.064633	0.094347	0.004031	0.094261	0	1000/1000
0.1	0.05	0.061741	0.092370	0.004332	0.092269	0	1000/1000
0.1	0.025	0.061471	0.091660	0.004501	0.091550	0	1000/1000
0.1	0.0125	0.059809	0.090348	0.004666	0.090228	0	1000/1000
0.02	0.2	0.013743	0.020376	0.000124	0.020375	0	1000/1000
0.02	0.1	0.012755	0.019077	0.000290	0.019074	0	1000/1000
0.02	0.05	0.012685	0.018632	0.000301	0.018629	0	1000/1000
0.02	0.025	0.012729	0.018453	0.000318	0.018450	0	1000/1000
0.02	0.0125	0.012482	0.018177	0.000320	0.018174	0	1000/1000
0.005	0.2	0.003440	0.005103	0.000014	0.005103	0	1000/1000
0.005	0.1	0.003208	0.004781	0.000048	0.004781	0	1000/1000
0.005	0.05	0.003179	0.004668	0.000048	0.004667	0	1000/1000
0.005	0.025	0.003167	0.004621	0.000052	0.004621	0	1000/1000
0.005	0.0125	0.003124	0.004551	0.000050	0.004551	0	1000/1000

The fixed-window and generating-parameter-test cells also retain 1000 paths each. Across these cells 5 final estimates are clipped; individual counts and all unbounded estimates are preserved in the numerical report and raw rows. This clipping changes only the bounded-risk summaries; no path is removed.

For Van der Pol the parameter pilot box is $[0.25, 2] \times [-2, -0.25] \times [0.1, 2] \times [-2, -0.1]$. DOP853 integration of its fitted model uses relative tolerance 10^{-9} and absolute tolerance 10^{-11}. The fitted trajectory is fixed during the coefficient-weight iterations. Tests and the basis transformation are fixed independently of observations. The relative covariance relaxation is scaled by the mean covariance diagonal, so common noise-variance scaling does not change GLS weights. One hundred iterations and tolerance 10^{-7} are computation settings, not a convergence theorem. Final coefficients are not clipped, and every finite estimate is included. Basis dimensions and both covariance matrices are preserved in the deterministic geometry output.

A five-iteration execution on the identical paths was retained as a stopping control. After inspecting its convergence diagnostics, the limit was increased to 100; this computational adjustment was not fixed before the five-iteration output. It does not add independent paths. The median relative coefficient change is 0.00e+00 and the maximum is 2.53e-03. No path is removed because of the

stopping test.

Table S2. Van der Pol finite-cohort decomposition and stopping diagnostics. Counts use 128 paths per row; attaining the iteration cap without convergence is retained in the sample. These empirical moments do not establish the uniform-integrability assumption of Corollary 1.

Noise	Windows	Weights	Bias	Dispersion	RMSE	Converged	Pilot clips
0.5%	30	iid	0.000312	0.002011	0.002035	128/128	0
0.5%	30	ou	0.000266	0.001983	0.002001	128/128	0
0.5%	120	iid	0.000188	0.001761	0.001771	128/128	0
0.5%	120	ou	0.000103	0.001727	0.001730	128/128	0
0.5%	480	iid	0.000181	0.001766	0.001775	128/128	0
0.5%	480	ou	0.000109	0.001734	0.001737	128/128	0
5%	30	iid	0.010013	0.025550	0.027442	128/128	0
5%	30	ou	0.009004	0.025569	0.027108	128/128	0
5%	120	iid	0.008493	0.024022	0.025479	128/128	0
5%	120	ou	0.007695	0.024258	0.025450	128/128	0
5%	480	iid	0.008464	0.024063	0.025509	128/128	0
5%	480	ou	0.008143	0.024253	0.025583	128/128	0

The raw scalar, geometry and Van der Pol outputs, including coefficient vectors, are archived. No failed stochastic fit is omitted; completion and expected seed counts are checked separately from stopping tolerances.

S9. Supporting numerical validation

S9.1. Full-record information and scalar likelihood

For $a = -1$, $x_0 = 2$, $T = 4$, $\tau = 0.4$ and $\sigma = 0.1$, continuous Fisher information is 144.187. The chosen scalar weak function has linearized information efficiency 0.063; Section S9.8 evaluates optimal combinations within declared window families. On 1,000 paired paths, the finest-grid median absolute errors were 0.2271 (95% interval [0.2132, 0.2454]) for the weak ratio and 0.0561 ([0.0533, 0.0603]) for bounded Gaussian likelihood. The latter uses known covariance and initial condition. Its finite-sample bias and boundedness prevent interpreting its MSE as an unbiased Cramér–Rao efficiency measure.

The coarsest scalar grid already captures 99.937% of continuous-record information. Extending the decay record from $T = 4$ to 8 adds only 0.564%, whereas a second independent record doubles information. This is a quantitative acquisition comparison under the stated known-covariance model, rather than a universal cost-optimal allocation.

Initial-state annihilation raises the weak influence variance lower bound from 0.006935 to 0.008213, an increase of 18.4%. The chosen window's efficiency relative to the nuisance-adjusted information reference is 0.075, compared with 0.063 relative to known-initial-state information. This separates the weak class restriction from additional window inefficiency. Deterministic quadrature verifies both influence identities to 10^{-9}; it adds no Monte Carlo paths.

S9.2. Fixed-window sparse regression

The fixed-window comparison contains 24,576 fits: four systems, two noise regimes, two amplitudes, four grids, three methods and 128 paths per condition. Thirty fourth-power polynomial bump windows remain fixed as the grid changes. For decay, $T = 4$ and $L = 1.6$; for oscillators, $T = 20$ and $L = 3$. Correlation ratio is $\tau/L = 0.1$. Relative MSTLS uses 50 thresholds between 10^{-4} and

one, loss weight 0.8 and no ridge penalty; raw and known-variance corrected features share these settings. Settings and seed ranges were fixed before outputs; this is not a public preregistration. Initial states and libraries are specified in the supplement.

Table S3. Corrected relative MSTLS at 5% noise and $h = 0.005$. Error and exact support are separate endpoints; 128 paths enter each row.

System	Error model	Median relative error	95% median interval	Exact support	95% support interval
Exponential decay	OU	0.0213	[0.0160, 0.0248]	128/128	[0.971, 1.000]
Exponential decay	Independent	0.0026	[0.0021, 0.0034]	128/128	[0.971, 1.000]
Damped oscillator	OU	0.0131	[0.0114, 0.0152]	128/128	[0.971, 1.000]
Damped oscillator	Independent	0.0013	[0.0012, 0.0016]	128/128	[0.971, 1.000]
Van der Pol	OU	0.0536	[0.0402, 0.0658]	81/128	[0.547, 0.711]
Van der Pol	Independent	0.0031	[0.0028, 0.0034]	126/128	[0.945, 0.996]
Duffing	OU	0.0804	[0.0698, 0.0990]	63/128	[0.407, 0.578]
Duffing	Independent	0.0061	[0.0053, 0.0077]	128/128	[0.971, 1.000]

In the Van der Pol OU condition, correct-support paths had median error 0.0340, whereas incorrect-support paths had median 0.7189. The aggregate median therefore combines parameter-estimation error with structure-selection error and cannot be read as an intrinsic information limit.

Intervals resample independent paths and retain paired mesh comparisons. Conditions share random-number indices and are not pooled as independent experiments. Exact-support intervals use Wilson limits. No multiplicity-adjusted global support claim is made. No forecast endpoint is included in this validation.

For corrected MSTLS at 5% OU noise, 3 of four system-specific paired median-improvement intervals for $h = 0.01$ to 0.005 lie wholly inside the $\pm 5\%$ band. This is a descriptive equivalence check of coefficient medians and does not guarantee support recovery.

S9.3. Coefficient covariance and Monte Carlo calibration

Joint coefficient covariance was tested on 2,000 Van der Pol paths at each amplitude, on known support. Actual errors subtract the clean discrete fit; a guard retains designs whose smallest support singular value is at least half its clean value. Every simulated path passed: the guard did not filter these samples, so guarded summaries equal the full empirical summaries. This observation does not establish finite unconditional estimator moments or control unobserved near-singular designs. The guard defines the intended validation region; it is not evidence that the linearization is accurate.

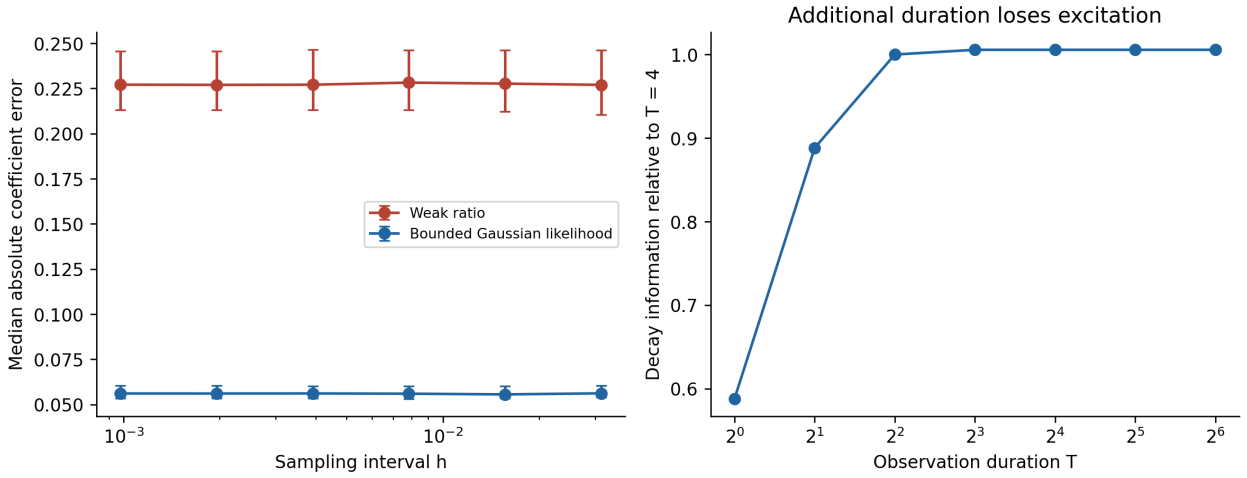


Figure S1: Scalar coefficient errors with median bootstrap intervals, and continuous Fisher information as duration increases.

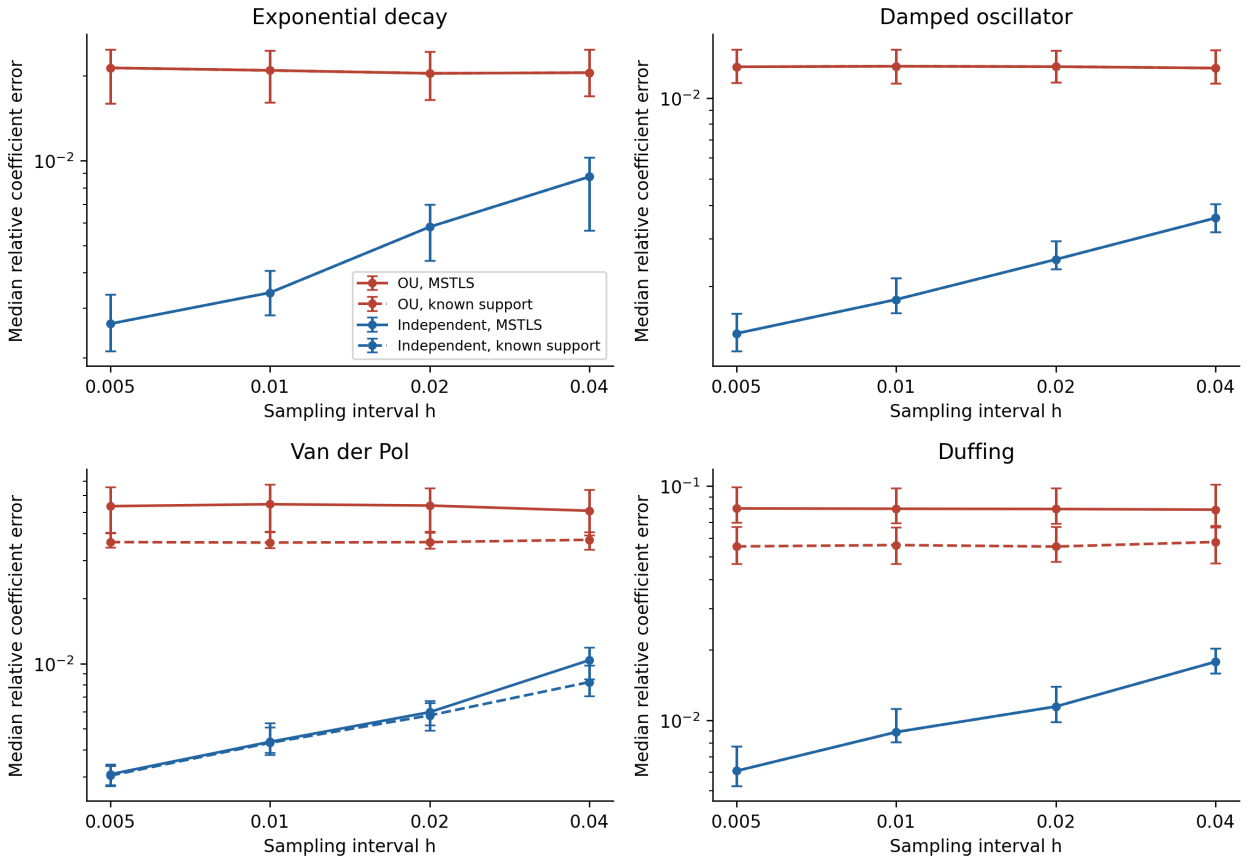


Figure S2: Median coefficient errors with 95% path bootstrap intervals. Known-support fits separate parameter estimation from sparse selection.

Table S4. Relative Frobenius covariance discrepancies and paired relative RMS linearization errors on known support.

Noise amplitude	Guarded paths	Linear covariance discrepancy	Guarded actual discrepancy	Paired linearization RMS
0.1%	2000/2000	0.040	0.040	0.025
1.0%	2000/2000	0.040	0.041	0.245
5.0%	2000/2000	0.040	0.294	1.156
20.0%	2000/2000	0.040	0.481	2.591

The linear covariance discrepancy is close to its Monte Carlo RMS floor at every amplitude because the standardized linear experiment is paired across amplitudes. Actual covariance agrees at 0.1% and 1% noise but has substantially larger discrepancy at 5% and 20%. Paired pathwise linearization already has appreciable error at 1%. Thus the coefficient surrogate is a local small-noise description, and the residual covariance theorem alone cannot validate it at the moderate-noise benchmark setting.

The last column is $R = \{\sum_m \|e_m - \ell_m\|^2 / \sum_m \|\ell_m\|^2\}^{1/2}$, where e_m is the actual coefficient displacement from the clean fit and ℓ_m its first-order surrogate. It is not a relative error of the estimated coefficients and is not constrained to $[0, 1]$. At 5% and 20% noise, omitted nonlinear terms and random-design coupling exceed the RMS magnitude of the first-order surrogate. A stable singular value alone does not suppress those terms.

In the 4,000-path nonlinear residual experiment, higher orders contributed 20.3% of corrected trace at 50% noise. The covariance calibration places the observed Frobenius discrepancy against the fourth-moment Monte Carlo scale and an affine-null bootstrap. The null procedure retains empirical non-Gaussian shape and forces the analytical covariance; its range is approximate, and its exceedance fraction is not an exact p-value.

For the corrected 50% condition, observed matrix discrepancy was 0.064; estimated fourth-moment relative RMS floor was 0.063, and the approximate null 95% range was [0.053, 0.072]. The 6.4% discrepancy therefore must be read relative to this uncertainty, rather than as an isolated accuracy percentage.

S9.4. Different initial conditions

A separate frozen experiment used 400 blocks with up to 16 independent Van der Pol recordings. Initial offsets were uniform on $[-0.25, 0.25]^2$ about $(2, 0)$; sensor noise covariance remained fixed. All methods supplied true support.

Table S5. Known-support aggregation with different initial conditions; 400 independent blocks per row.

Method	M	Median relative error	95% median interval
Stacked design	1	0.03577	[0.03305, 0.03888]
Stacked design	16	0.03785	[0.03684, 0.03857]
Mean weak matrices	1	0.03577	[0.03305, 0.03888]
Mean weak matrices	16	0.00735	[0.00693, 0.00773]
Mean sensor path	1	0.03577	[0.03305, 0.03888]
Mean sensor path	16	0.10847	[0.10523, 0.11193]

Mean weak matrices improve with independent acquisition in this heterogeneous setting, whereas stacking retains a stable error and sensor-path averaging deteriorates. The latter estimates the dynamics of an averaged path that generally fails the original nonlinear equation. The evidence

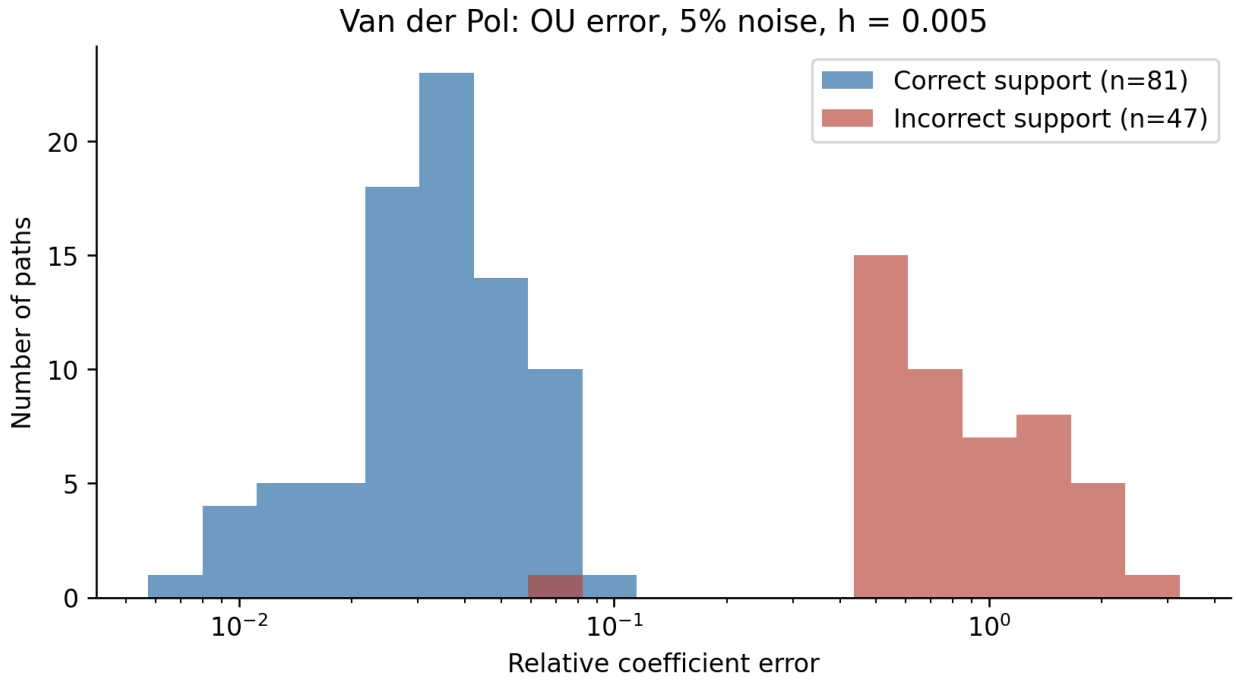


Figure S3: Error distributions conditional on correct or incorrect selected support in the finest-grid Van der Pol OU condition. Counts accompany each distribution; bimodality is not presumed.

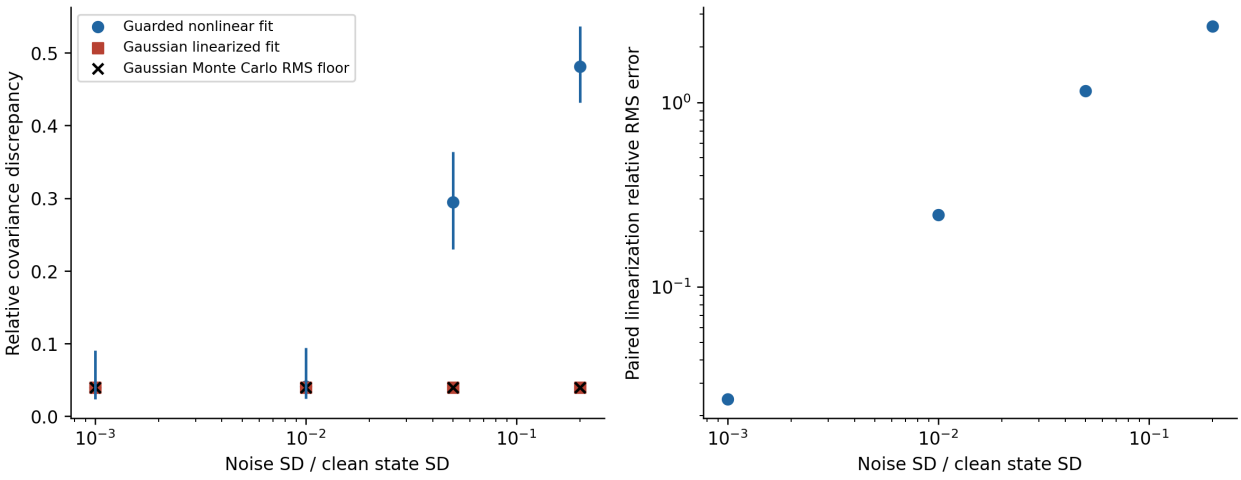


Figure S4: Joint coefficient covariance discrepancies, bootstrap intervals for guarded actual covariance, Gaussian Monte Carlo RMS references, and paired pathwise linearization error.

supports a distinction between aggregation operations, under the supplied support and excitation used here. Coefficient-level bias and dispersion are reported in Section S9.9; the near-equal single-record and stacked median errors alone do not identify the bias fraction.

S9.5. Covariance reweighting component

An additional comparison supplies known support and compares ordinary raw-feature least squares, the iid covariance IRLS step from WENDy Algorithm 2, and its known-OU covariance extension. It retains our fixed windows and uses ten iterations, relative stopping tolerance 10^{-6}, known coordinate variances and scale-aware covariance relaxation 10^{-10}. These choices make it a component experiment rather than a reproduction of complete WENDy.

Table S6. Van der Pol covariance-reweighting components: parameter one, initial state (2,0), T=20, 5% state-scaled noise, OU correlation time 0.3, 30 fixed windows of full width three, and 128 paths per condition. True support is supplied.

Error model	Component	h	Median relative error	95% median interval	Converged
OU	OLS, known support	0.04	0.0322	[0.0289, 0.0378]	n/a
OU	OLS, known support	0.005	0.0325	[0.0289, 0.0363]	n/a
OU	WENDy iid IRLS core	0.04	0.0185	[0.0164, 0.0205]	126/128
OU	WENDy iid IRLS core	0.005	0.0186	[0.0165, 0.0203]	127/128
OU	OU IRLS core	0.04	0.0184	[0.0163, 0.0202]	124/128
OU	OU IRLS core	0.005	0.0181	[0.0161, 0.0198]	121/128
Independent	OLS, known support	0.04	0.0080	[0.0071, 0.0092]	n/a
Independent	OLS, known support	0.005	0.0034	[0.0031, 0.0037]	n/a
Independent	WENDy iid IRLS core	0.04	0.0057	[0.0047, 0.0065]	128/128
Independent	WENDy iid IRLS core	0.005	0.0032	[0.0030, 0.0034]	128/128
Independent	OU IRLS core	0.04	0.0058	[0.0051, 0.0065]	128/128
Independent	OU IRLS core	0.005	0.0032	[0.0029, 0.0035]	128/128

At h=0.005 under OU error, the median decreases from 0.0325 for ordinary least squares to 0.0186 for iid IRLS and 0.0181 for OU IRLS. Reweighting therefore improves this component comparison, but its two IRLS medians are close and their displayed intervals overlap; the table alone does not establish superiority of the OU extension. OU errors remain nearly unchanged across the two meshes, whereas independent-noise errors decrease. Known covariance and supplied support distinguish this comparison from the complete adaptive pipelines in Section S9.6. All iteration-capped estimates enter the summaries. The convergence column reports the prescribed tolerance; it does not assess model correctness.

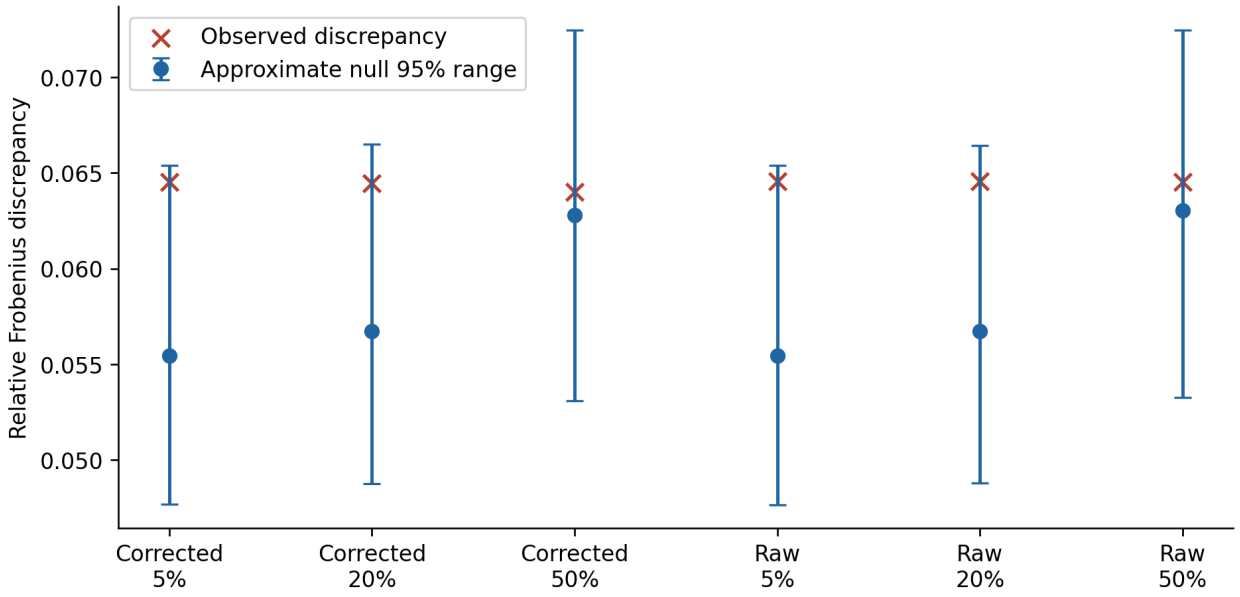


Figure S5: Observed covariance discrepancies and approximate covariance-null bootstrap ranges for three discrete amplitudes and two correction choices.

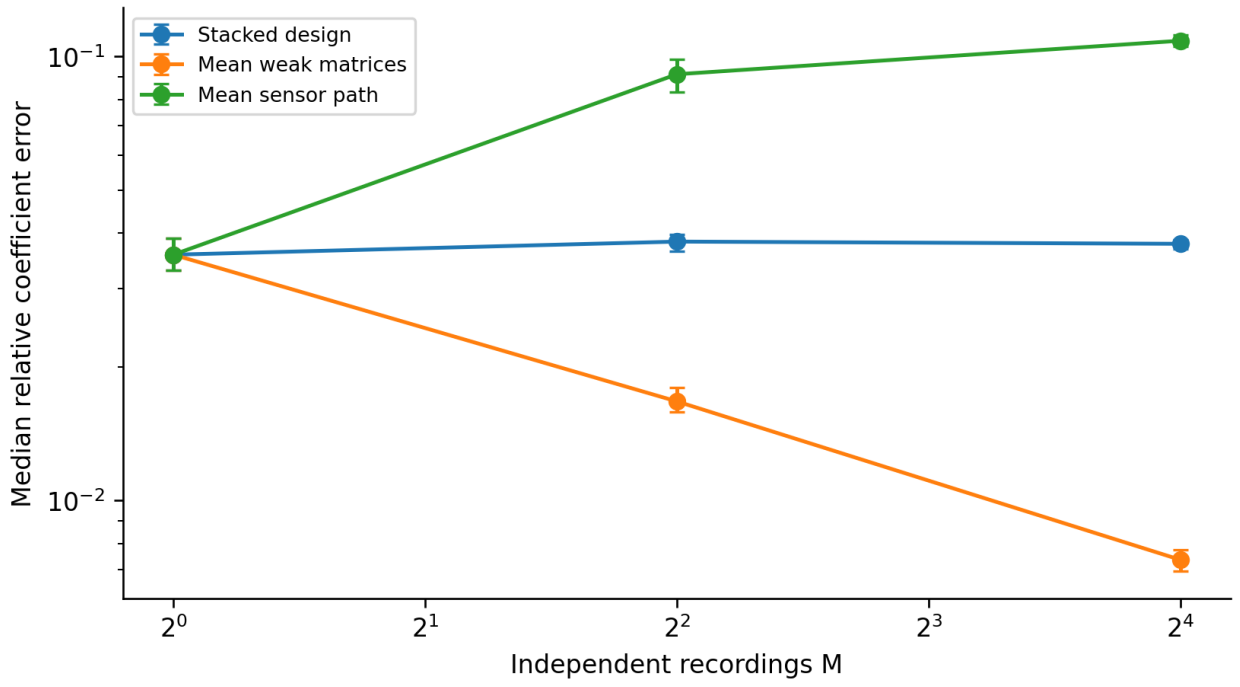


Figure S6: Independent recordings with different initial states. Mean weak matrices and sensor-path averaging define different estimating equations.

S9.6. Complete adaptive author pipelines

A separate frozen comparison executes the complete pinned author WSINDy, WENDy and WmSINDy algorithms [4–6]. WSINDy selects its test-function widths spectrally, places centers by the gradient, applies automatic SMAF smoothing and library scaling, and selects the MSTLS threshold. WENDy estimates noise, selects four adaptive radii and an SVD basis, and runs its full covariance iteration and stopping rules. WmSINDy jointly learns coefficients and noise with forward/backward RK4 and weak-residual losses, updates windows from the denoised data, and prunes and refits coefficients after each of eight cycles of 5000 Adam steps. Versions and compatibility adapters are detailed in Supplement S6.

The benchmark uses Van der Pol, five-percent state-scaled noise, fixed duration 20, correlation time 0.3, and 16 independent paths per condition. Meshes 0.04 and 0.01 are nested. All methods receive identical observations verified by SHA256; discovery methods share the nine nonconstant monomials of degrees one through three. Fixed-window MSTLS is rerun on this library. WENDy receives the prespecified correct four-term structure and is assessed as parameter estimation. Hermite MSTLS uses the known calibration variance, whereas author pipelines estimate or reconstruct noise from observations.

Table S7. Discovery pipelines at h=0.01; 16 independent records per row. Median intervals are path bootstraps. Support counts include all planned fits.

Error model	Method	Median error [95% interval]	Exact support	Failed fits
OU	Fixed raw MSTLS	0.0402 [0.0300, 0.5657]	11/16	0/16
OU	Fixed Hermite MSTLS	0.0420 [0.0311, 0.5656]	11/16	0/16
OU	Adaptive author WSINDy	0.0430 [0.0354, 0.0536]	14/16	0/16
OU	Adaptive author WmSINDy	0.0153 [0.0110, 0.0180]	16/16	0/16
Independent	Fixed raw MSTLS	0.0043 [0.0036, 0.0059]	16/16	0/16
Independent	Fixed Hermite MSTLS	0.0029 [0.0023, 0.0046]	16/16	0/16
Independent	Adaptive author WSINDy	0.0086 [0.0070, 0.0118]	16/16	0/16
Independent	Adaptive author WmSINDy	0.0059 [0.0036, 0.5396]	10/16	0/16

Table S8. Parameter estimation with supplied support at h=0.01; 16 independent records per row.

Error model	Method	Median error [95% interval]	Failed fits
OU	Adaptive author WENDy (known support)	0.0151 [0.0120, 0.0212]	0/16
OU	Fixed raw, known support	0.0300 [0.0224, 0.0367]	0/16
OU	Fixed Hermite, known support	0.0312 [0.0235, 0.0384]	0/16
Independent	Adaptive author WENDy (known support)	0.0029 [0.0023, 0.0035]	0/16
Independent	Fixed raw, known support	0.0043 [0.0036, 0.0058]	0/16
Independent	Fixed Hermite, known support	0.0029 [0.0023, 0.0042]	0/16

On the finer OU grid, WmSINDy recovered support on 16/16 paths with median error 0.0153, whereas adaptive WSINDy recovered 14/16 with median 0.0430. The paired median error difference, WmSINDy minus WSINDy, was -0.0279 with a descriptive 95% bootstrap interval [-0.0378, -0.0204]. Under independent errors, WmSINDy recovered support on 10/16 paths, compared with 16/16 for fixed Hermite MSTLS. Its wide median interval reflects selection failures despite the full training budget. Thus the favorable OU result does not establish uniform superiority across observation regimes. These unadjusted comparisons are descriptive and limited to the frozen example.

The supplied-support comparison likewise concerns finite-sample accuracy: adaptive WENDy improves the OU coefficient median relative to these fixed-window fits, although its released noise model assumes independent observations. This empirical behavior does not establish calibrated uncertainty under OU error. Better window selection or reconstruction can improve accuracy within the finite information budget; it does not contradict the full-path infill bound.

All 448 planned fits are archived; 0 computational failures occurred. Error medians condition on completed fits. Wilson intervals, coarse-grid values, and paired mesh differences are retained in the reproducibility report. With 16 paths, support fractions have substantial binomial uncertainty. These conditions compare complete algorithms on one oscillator and do not establish a general ranking or an asymptotic plateau.

S9.7. Van der Pol information reference

The OU experiment in Section S9.6 has $x' = av$ and $v' = bx + cv + dx^2 v$, with four independently free active coefficients $(a, b, c, d) = (1, -1, 1, -1)$. The initial state (2,0) is known. We hold the calibrated coordinate noise variances fixed when differentiating the mean, rather than re-estimating state-scaled variance at perturbed parameters. The calibration is exactly that of the benchmark fine grid. Coefficient and initial-state sensitivities solve $S' = JS + \partial_\theta f$ and $Z' = JZ$, with $S(0) = 0$ and $Z(0) = I$. OU innovations give discrete information; the continuous norm in Section 4.1 gives its limit.

For four active coefficients, the relative RMS reference is $\{\mathrm{tr}(\mathcal{J}_{\theta\theta}^{-1})\}^{1/2}/\|\theta\|_2$. The weak initial-state-annihilation reference replaces information by its Schur complement. Both are local covariance bounds, rather than bounds on error medians or on every biased estimator.

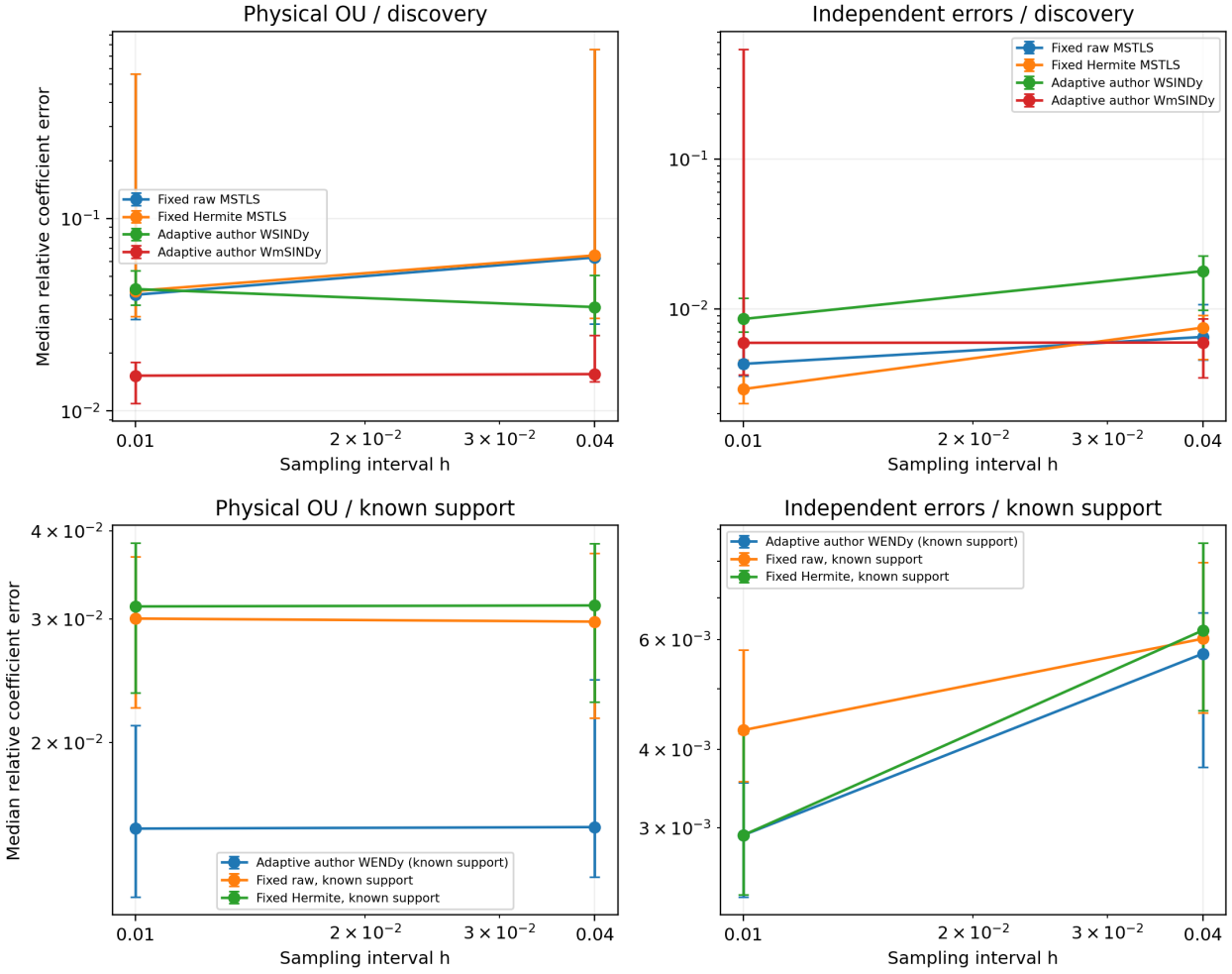


Figure S7: Complete adaptive author pipelines and matched fixed-window comparisons. Error bars are 95% path bootstrap intervals with n=16 per cell. WENDy uses known support; the discovery methods do not.

Table S9. Van der Pol OU relative RMS information references for the exact Section S9.6 calibration. Four active coefficients are independently free. The second column assumes known initial state; the third also imposes weak initial-sensitivity annihilation.

Observation	Known-initial reference	Weak influence reference
Grid h=0.04	0.013115	0.016929
Grid h=0.01	0.013094	0.016906
Continuous record	0.013093	0.016904

For the same 16 OU records, WmSINDy has relative RMSE 0.01926 (descriptive 95% bootstrap interval [0.01324, 0.02576]) and WENDy 0.02154 ([0.01446, 0.02927]). Their RMSEs are respectively 1.47 and 1.65 times the known-initial local reference. This scale comparison is more informative than comparing their medians near 0.015 with a variance bound. It is not an efficiency certificate: sparse selection and nonlinear bias are present, the sample is small, and the regular locally unbiased assumption is unverified. WmSINDy also includes trajectory-prediction terms, so it is not restricted to boundary-free weak influences.

A one-parameter Van der Pol model fixes a=1 and b=-1 and ties c=mu, d=-mu. Its continuous standard-deviation reference for mu is 0.002492. That substantially stronger constraint changes the estimation problem and must not be used to judge four-coefficient discovery. Central finite differences check the four sensitivities with relative discrepancies below 1.6e-07; refining information quadrature from h=0.0025 to 0.00125 checks the reported reference.

S9.8. Optimal efficiency within declared window spans

The scalar parameters are those of Section S9.1. We optimize the covariance-weighted linear combination in Section 4.6, including every cross-window covariance. Family F30 contains 30 fourth-power bump windows of full width 1.6; centers range uniformly from 0.805 to 3.195. Family F120 contains 60 centers for each of widths 1.6 and 3.0, with centers 0.005 inside the admissible support range. These are declared finite families, rather than an exhaustive search over all shapes and widths. The reference single window has width three and center two.

Table S10. Oracle optimal local influence variance and efficiency within each scalar window span. Information references are 144.187 for known initial state and 121.752 after initial-state projection. The table uses 8192 quadrature intervals and relative covariance-eigenvalue cutoff 1e-12.

Family	Retained rank	Variance	Known-initial efficiency	Weak-class efficiency
Single window	1	0.109776	0.0632	0.0748
F30 (30 windows)	30	0.010656	0.6508	0.7708
F120 (120 windows)	68	0.009339	0.7426	0.8795

Thus 0.063 describes one window, not the optimal efficiency of the weak family. F120 attains 0.743 of known-initial information in the retained span. Tightening the cutoff from 1e-10 to 1e-12 raises retained rank from 41 to 68; at 1e-10 its known-initial efficiency is 0.718. This sensitivity is disclosed rather than treated as an exact infinite-dimensional optimum. Doubling quadrature intervals changes every reported variance by less than 0.0004%. All optimal influences satisfy coefficient normalization and initial-state annihilation numerically. Their efficiencies remain below the initial-state-projected ceiling.

S9.9. Bias and dispersion of aggregation

The coefficient-preserving execution replays exactly the 400 independent blocks in Table S5; it adds no independent samples. For normalized coefficient errors, the empirical identity states that squared RMSE equals squared bias plus squared dispersion. Table S11 reports the norm of the sample mean error as bias and the square root of the second term as dispersion. Bias here is an empirical estimate, not a proved population limit. We also subtract sample-mean sampling variance when estimating the squared-bias fraction; the adjustment is small for 400 blocks.

Table S11. Relative coefficient bias, dispersion and RMS error for heterogeneous Van der Pol aggregation; the same 400 blocks as Table S5. Values are dimensionless. Bias fraction uses the unadjusted empirical squared mean error.

Method	M	Bias	Dispersion	RMSE	Squared-bias fraction
Stacked design	1	0.02967	0.03229	0.04385	45.8%
Stacked design	16	0.03738	0.00835	0.03830	95.3%
Mean weak matrices	1	0.02967	0.03229	0.04385	45.8%
Mean weak matrices	16	0.00239	0.00893	0.00924	6.7%
Mean sensor path	1	0.02967	0.03229	0.04385	45.8%
Mean sensor path	16	0.11044	0.02914	0.11422	93.5%

For stacking, dispersion decreases from 0.03229 at M=1 to 0.00835 at M=16, while bias changes from 0.02967 to 0.03738. The estimated bias contribution is 45.8% of single-record MSE and 95.3% at M=16; the sample-mean-adjusted fractions are 45.6% and 95.2%. The stacking floor is therefore dominated by systematic random-design error as acquisition increases. Whether the single-record error is mainly bias must be judged by this decomposition, not by the two median norms. Mean weak matrices reduce both errors under the stated excitation, whereas sensor-path averaging produces nonlinear averaging bias.

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